Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

EFRE
BAYERN 2021-2027

DATA DRIVEN
DECISIONS

Smart Mobility Würzburg

Planning Public Transport to Match Demand with Data and AI

Master Project Seminar: Organization, Roles, and Phases

Winter Semester 2026/27



Our Working Methodology

Two Phases:

- **Phase 1: Individual Research** – Each student writes a term paper (10–15 pages) to build deep expertise in their domain.
- **Phase 2: Collaborative Implementation** – The team comes together to build the integrated system.

The “Domain Owner” Concept:

- No strict silos – every team member supports each other during implementation.
- Each student **owns** one domain and is responsible for its success.
- Ownership means: driving decisions, defining interfaces, and ensuring quality.

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Overview of the 3 Domains

1. **Data Engineering & Infrastructure**
   Ingestion, storage, and delivery of all mobility data.
2. **Traffic Simulation (Digital Network)**
   Modeling and simulating the Würzburg transit network.
3. **Capacity Planning & Predictive Analytics**
   Forecasting demand and optimizing schedules with ML.

Shared by everyone: Visualizing the results and what was accomplished.

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Domain 1: Data Engineering

Core Tasks for the Term Paper:

- Evaluate data sources and formats (e.g., GTFS, real-time APIs).
- Compare spatial database architectures (e.g., PostGIS vs. NoSQL).
- Design robust ETL pipelines for continuous data processing.
- Define the **API contract**: What data do I need, and what do I provide to the team?

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Domain 2: Traffic Simulation

Core Tasks for the Term Paper:

- Research and compare different traffic simulation approaches.
- Evaluate open-source frameworks (e.g., SUMO, MATSim).
- Develop a concept for importing the real Würzburg road/bus network into the simulation.
- Define the **API contract**: What data do I need, and what do I provide to the team?

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Domain 3: Capacity Planning & ML

Core Tasks for the Term Paper:

- Compare time-series forecasting algorithms (statistical vs. ML).
- Explore Operations Research methods for schedule and capacity optimization.
- Define necessary feature engineering (e.g., weather, holidays, traffic conditions).
- Define the **API contract**: What data do I need, and what do I provide to the team?

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Shared Task: Visualizing the Results

Everyone on the team contributes:

- Each domain owner visualizes the results of their own domain.
- Together, the team builds a **dashboard** that presents the results.
- The dashboard should be **interactive** wherever possible.
- Its content is driven by the domains and the questions each of them raises.

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AI-Assisted Development

The New Reality:

- We know you use LLMs and coding agents – **and we expect you to.**
- It has never been easier to turn ideas into working code.
- But: it has also never been easier to produce **unmaintainable, undebuggable garbage.**

Coding Agents You Should Know:

- **GitHub Copilot** – Inline suggestions, integrated in VS Code & JetBrains.
- **Cursor / Windsurf** – AI-native IDEs with multi-file editing and chat.
- **Claude Code** – CLI agent with deep codebase understanding.

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Think First, Then Code

Before You Write a Single Line:

- **Write a specification** – What are you building? Why? How?
- **Define interfaces and data contracts** – especially across domains.
- **Set up your project** – structure, linting, formatting, CI, tests.
- **Give the agent context** – `CLAUDE.md`, `.cursorrules`, or Copilot instructions.

Spec-Driven Development:

- **GitHub Spec Kit** – Write specs first, then let the agent implement.
  Gated phases: Specification → Design → Implementation → Validation.
- **Rule:** The agent's output is **your responsibility.** Review, test, understand every line.

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Next Steps

- Create a team (there will be two teams).
- Create a communication channel and invite the supervisors.
  Suggestions: Slack, Discord, …
- Decide who is responsible for which domain.
- Create a project sketch and define milestones.
  Suggestion: GV-Sprints – Video | The Sprint Book
- Organize regular feedback meetings with the supervisors.
- **Presentation: Friday, November 20, 2026** (planned in person).
- **Deadline for the term paper: Sunday, November 22, 2026.**

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